

GY111: Physical Geology

Spring 2025

Instructor

Dr. Benjamin J. Linzmeier

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Preferred contact: Canvas messages

Office Hours: 11 AM – 12 PM Mo, W, F
2 PM – 5 PM M

Office: Life Science Building room 344

Correspondence policy: Within 24 hours on weekdays; 48 hours on weekends

Class information

Classroom: Mitchell Center Room 1101

Class Hours: 9:05 AM - 9:55 AM Mo We Fr

Course description

How does the Earth work as a system? In this class we will learn the basics of the science of Geology with a focus on physical processes and materials. We will cover major topics including: plate tectonics, mineral chemistry, the rock cycle, geological time, sedimentary processes, climate change, and geological map reading.

Important Dates

January 20th - Last day to drop without WD

February 28th - In person midterm exam

April 11th - Last day to withdraw

May 7th, 8-10 AM - In person final exam

See the [Academic Calendar](#) for more information.

Course goals

1. Explain and apply the scientific method as it is practiced in the Geosciences
2. Link observations of the Earth on multiple scales to the development of theories
3. Explain how the hydrosphere, biosphere, and geosphere interact on multiple timescales and predict how changes in one can affect others
4. Read and interpret topographic and geological maps
5. Identify common minerals and rocks
6. Link geological materials to Earth processes
7. Relate facts about the Earth system to human problems

Class textbook

We will be using the e-book of Exploring Geology by Stephen Reynolds and Julia Johnson. I chose this book because it has guided practice with readings and includes interactive web-based activities based on societally relevant problems related to Geology.

Lectures and papers

Both lecture slides and supplemental readings will be available via Canvas. Slides will generally be posted after lecture.

Date	Class	Week	Lecture
2025-01-13	1	1	Introduction and overview
2025-01-15	2	1	Nature of Geology
2025-01-17	3	1	Geology as a science
2025-01-22	4	2	Geological time
2025-01-24	5	2	Geological time
2025-01-27	6	3	Continental drift
2025-01-29	7	3	Plate tectonics
2025-01-31	8	3	Earth's structure
2025-02-03	9	4	Earthquakes
2025-02-05	10	4	Earthquakes
2025-02-07	11	4	Earthquakes
2025-02-10	12	5	Minerals
2025-02-12	13	5	Minerals
2025-02-14	14	5	Minerals
2025-02-17	15	6	The Rock cycle
2025-02-19	16	6	Igneous Rocks
2025-02-21	17	6	Igneous Rocks
2025-02-24	18	7	Volcanoes
2025-02-26	19	7	Volcanoes
2025-02-28	20	7	Midterm
2025-03-10	21	9	Metamorphism
2025-03-12	22	9	Deformation and Landscapes
2025-03-14	23	9	Deformation
2025-03-17	24	10	Weathering
2025-03-19	25	10	Erosion
2025-03-21	26	10	Sedimentary Rocks
2025-03-24	27	11	Planetary Geology
2025-03-26	28	11	Geobiology
2025-03-28	29	11	Geobiology
2025-03-31	30	12	Energy and Mineral Resources
2025-04-02	31	12	Energy and Mineral Resources
2025-04-04	32	12	Energy and Mineral Resources
2025-04-07	33	13	Groundwater
2025-04-09	34	13	Groundwater
2025-04-11	35	13	Groundwater
2025-04-14	36	14	Streams and Flooding
2025-04-16	37	14	Streams and Flooding
2025-04-18	38	14	Streams and Flooding
2025-04-21	39	15	The Climate System
2025-04-23	40	15	The Climate System
2025-04-25	41	15	The Climate System
2025-04-28	42	16	Coasts, Glaciers, and Changing sea level
2025-04-30	43	16	Coasts, Glaciers, and Changing sea level
2025-05-02	44	16	Coasts, Glaciers, and Changing sea level

Student conduct

Attendance policy

- Students who complete less than 50% of assignments/exams will be assigned a failing (F) grade.
- Review the 'Attendance and Absences Policy', in the Undergraduate and Graduate Bulletin for attendance and absences policy

Grading

Missing work

Everyone will have their lowest two discussion posts, readings, and quizzes dropped. This means that you have two weeks of built-in flexibility for life events that come up (e.g. illness, unexpected travel).

Late work policy

If you need additional flexibility based on difficulties arising from life, please let me know. That said, given dropping the lowest two weeks worth of work means that I will not be allowing for late quizzes or discussion posts.

Extra credit

Extra credit may be made available throughout the semester, but readings, quizzes, and discussion posts .

Assignments will be returned within a week of submission.

Grade	Range
A	90-100
B	80-89
C	70-79
D	60-69
F	< 60

Assignment weighting will follow:

Item	Weight
Discussion posts	~13 %
Participation	~13 %
Readings	~39 %
Quizzes	~13 %
Midterm Exam	~7 %
Final Exam	~15 %

Academic disruption policy

The University of South Alabama's policy regarding Academic Disruption is found in The Lowdown, the student handbook. <http://www.southalabama.edu/lowdown/academicdisruption.shtml>. Disruptive academic behavior is defined as individual or group conduct that interrupts or interferes with any educational activity or environment, infringes upon the rights and privileges of others, results in or threatens the destruction of property and/or is otherwise prejudicial to the maintenance of order in an academic environment. At all times students will be cordial, courteous and respectful of faculty members and fellow students. Cell phones, videotaping, and other electronic devices are not allowed; however, you may use a laptop for note taking. If your laptop is used for other purposes, the instructor holds the right to revoke laptop use.

Academic honesty

The University of South Alabama's policy regarding Student Academic Conduct Policy is found in The Lowdown <http://www.southalabama.edu/lowdown/academicconductpolicy.shtml>: The University of South Alabama is a community of scholars in which the ideals of freedom of inquiry, freedom of thought, freedom of expression, and freedom of the individual are sustained. The University is committed to supporting the exercise of any right guaranteed to individuals by the Constitution and the Code of Alabama and to educating students relative to their responsibilities.

Violation of academic conduct policy may result in receiving 0 credit for the affected exam/assignment.

Do not pass others work off as your own. This constitutes plagiarism and seriously undermines your education. Students may learn about the meaning of plagiarism and how to avoid it at the following link: <http://www.southalabama.edu/univlib/instruction/plagiarismforstudents.html>

Course and Teacher evaluation

Student input for the purpose of course improvement is taken very seriously and will potentially be done periodically. Please take the time to evaluate this course and the instructor, especially at the end of the semester. Evaluations will in no way affect your grade.

Safety

Building Safety

Fire

Random fire safety drills may occur during the course of the term. You will be expected to evacuate the building and assemble at the designated location: outside by the northeast corner of the building (follow your instructor).

Tornado

Move away from external windows to the hallway in the basement, first, or second floor.

Active Shooter Incident

“Get Out” of the building as long as they can do so safely. Once you have exited the building and are at a safe location, “Call Out” to law enforcement using 911, USA Police at 460-6312, or press the Emergency button on your LiveSafe Mobile Safety App. If you can’t get out, “Shelter in Place” in a secure location that is away from windows and doors, preferably a location that can be locked and barricaded.

Campus resources

COVID policy

While on campus, students are required to follow all USA COVID-19 policies and practices. These policies can be found at <https://www.southalabama.edu/coronavirus/>. Be advised that policies may change during the semester, and students should check the University's Coronavirus Update page regularly.

There is also a student incentive program for getting vaccinated. Details about incentives can be found at <https://www.southalabama.edu/coronavirus/vaccineincentive/>

Writing center

Over the course of this semester, seek help at the Writing Center if you would like additional guidance and critique. You can access the Writing Center at [at their website](#). During graduate school, I found workshops and writing groups at the Writing Center on campus to be helpful in building my motivation and teaching me new strategies for writing efficiently.

Disabilities

In accordance with the Americans with Disabilities Act, students with bona fide disabilities will be afforded reasonable accommodations. The Office of Special Student Services (OSSS) will certify a disability and advise faculty members of reasonable accommodations. If you have a specific disability that qualifies you for academic accommodations, please notify the instructor/professor and provide certification from the Office of Special Student Services. OSSS is located at 5828 Old Shell Road at Jaguar Drive, (251-460-7212).

Food insecurity

If you are subject to food insecurity, JagPantry provides a food for students needing assistance. More information can be found [on their website](#)

Mental health

Being a university student is stressful and can negatively impact mental health. The University Counseling and Testing Center provides confidential, free counseling and crisis intervention services to eligible USA students and consultation and outreach services to members of the USA community. More information can be found [at their website](#).

Disaster plan

In case of Hurricane we will follow university recommendations and off-campus accommodations will be made. Our broad goals for the semester will continue but mode of delivery and content may change.

In case of a shift to remote learning, we will heavily rely on Canvas. We will have at least one synchronous lecture meeting that will be recorded and made available through Canvas. We will use digital options for rock samples and may also use [Rockd.org](https://rockd.org) for local outcrop locations if we are dispersed.

Final thoughts

This document is a roadmap for our semester. We learn about the Earth together and our individual experiences shape how we interpret and value data. Like all your classes, you will get out what you put into this course. Asking for help from one another and your instructors is important, don't be afraid to ask a question about something you don't know or if you want to check your knowledge about something you think you know.

If this document is updated, a copy will be supplied to you via Canvas and changes will be announced in class.